

How about more categories?

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 - **Contrasts**: Each category is compared to a **baseline**, and the coefficients are comparisons between baseline and levels
 - **One-hot**: coefficients are means of each level of the predictor
 - **Residuals**: an overall mean is measured, and coefficients are differences between each level and the global mean

```
> x = sample(LETTERS[1:3], 9, replace = TRUE)
> y = 1 + ifelse(x == "A", 0,
                ifelse(x == "B", 1, 2)) + rnorm(9)
> df = tibble(y, x)
> df
# A tibble: 9 × 2
      y x
  <dbl> <chr>
1  1.25 B
2  0.870 A
3 -0.00180 A
4  1.18 B
5  2.03 C
6  2.60 B
7  2.55 B
8  3.92 C
9  4.66 B
```

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Contrasts, the default in most `lm()` functions

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```
> fit_contrasts = stan_glm(y ~ x, data = df)
> summary(fit_contrasts)[1:3, 1:3]
```

	mean	mcse	sd
(Intercept)	0.72	0	0.13
xB	1.32	0	0.20
xC	2.20	0	0.186

Q: How do we get estimates for the mean in each class?